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Roll No

ME-8003 (3) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Advance Machining Process
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What do you understand by the term Non-traditional or unconventional machining methods?
How do you classify the common unconventional machining methods? 7
b) Define ultrasonic and describe the process in which these are used to machine the material. 7
2. a) Explain principle of water jet machining with diagram. 7
b) What are the specific advantages of using chemical machining over electrochemical machining? 7
3. a) Describe Electro Chemical Deburring process (ECD) and what are its advantages. 6
b) What is the principle of electrochemical machining? Explain with diagram. 8
4. a) How metal removal rate is calculated in ECM process? Explain. 7
b) What is laser and how it is used to machine the materials? 7
5. It is required to drill a hole of 1.69mm diameter to a depth of 7.32mm steel sheet using RC circuit by electro-discharge machining method using a brass electrode. Surface required is to be 20 microns. Determine the source voltage to be setup for a laser and resistance setting of 120 microfarads and 100 ohms respectively. Also find out the time required for drilling. Assume sparking period = 100 μ .sec. 14
6. a) Explain what is meant by non-transferred and transferred mode of plasma arc? What are the advantages of each? 10
b) Give examples of 3D printing. 4
7. a) What is Rapid Prototype? Provide a definition of the term. What are the three types of starting materials in rapid prototyping? 7
b) Describe the RP technology called laminated object manufacturing. 7
8. a) What are the three steps in the UGA process. 7
b) Define micro-electrochemical system. What is approximate size scale in micro system technology? 7
